WARDRIVE CYD-C5

Simple Wi-Fi wardriver for the NM-CYD-C5 (ESP32-C5, 2.8" ST7789 LCD with touch, SD card and GPS). Scans 2.4 GHz and/or 5 GHz Wi-Fi, shows GPS status on the display, and logs every network to the SD card in WiGLE CSV format. GPS: GPS+BDS Beidou dual-mode module (ATGM336H-compatible, NEO-M8N replacement).

1. What you need

- NM-CYD-C5 board with the 2.8" display and resistive touch
- microSD card, formatted as FAT32
- GPS+BDS Beidou dual-mode module (ATGM336H-compatible, NEO-M8N replacement). Outputs NMEA at 9600 baud by default. Needs an antenna with clear sky view.
- USB-C cable and a computer with Arduino IDE 2.x

2. Arduino IDE setup

- Install the **esp32 by Espressif Systems** board package, version 3.3.x. Select board **ESP32C5 Dev Module**.
- Install the library **GFX Library for Arduino** (by moononournation), version 1.6.9 or later.
- Install the library **XPT2046_Touchscreen** (by Paul Stoffregen).
- Set these board options under **Tools**:

Tools menu	Setting
USB CDC On Boot	Enabled
PSRAM	Disabled
Flash Size	16MB

Open *wardrive_cal.ino* and click Upload. Keep the board connected until the upload finishes.

3. Wiring and pins

Function	GPIO	Note
GPS RX (from GPS module TX)	8	Connect the GPS module TX pin here (header pin GPIO8).
GPS TX	-	Not used
SD card CS	10	Shares SPI bus with display and touch
LCD / touch / SD SPI	SCK 6, MISO 2, MOSI 7	One shared bus
LCD CS / DC / backlight	23 / 24 / 25	Backlight is on when HIGH
Touch CS	1	Touch IRQ not used

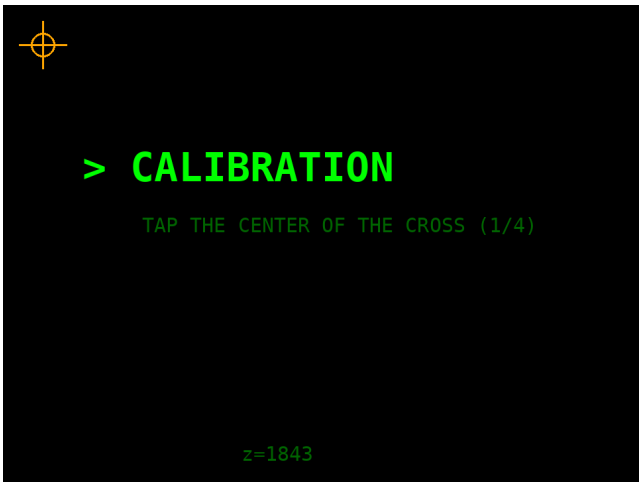
Power the GPS module from the 3.3 V and GND pins on the board header, next to GPIO8. Connect only the module's TX line to GPIO8. The module's RX line and the board's TX pin are not used.

4. First start and calibration

The touch screen must be calibrated once. On first boot the calibration screen appears automatically. The values are saved in the board's flash memory, so you only need to repeat this if touch stops working correctly.

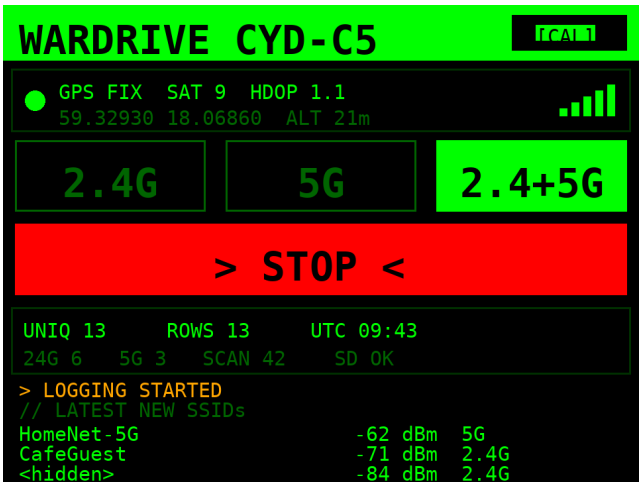
- A cross appears in each corner in turn: top left, top right, bottom right, bottom left.
- Tap the centre of each cross firmly and wait until the next one appears.

- When all four are done, the screen shows **CALIBRATED** and the main screen opens.
- If it shows **CALIBRATION FAILED**, the taps were too weak or too close together. The calibration starts over automatically.
- Start calibration again at any time with the **[CAL]** button in the top right corner.



Calibration screen. Tap the centre of each cross, starting top left. Simulated render, not a photo.

5. Main screen



Main screen while logging, with a 2.4+5G scan and GPS fix. Simulated render of the display layout, not a photo. Sample values only.

Area	What it shows
Header	Program name and the [CAL] button
GPS panel	Status dot: green = fix, yellow = searching, red = no data. Satellites, HDOP, coordinates and altitude. Bars show satellite count.
Band buttons	**2.4G**, **5G**, **2.4+5G**. The selected band is filled green. The choice is remembered after restart.
START / STOP	Starts or stops logging to a new file on the SD card
Statistics	UNIQ = unique networks this session, ROWS = lines written, UTC = GPS time, 24G / 5G = networks seen in the last scan, SCAN = scans done, SD = card status
Message line	Short status messages, for example band changes or errors
Latest SSIDs	The three most recent new networks found

6. Logging

- Press **START** to create a log file. The name is *wd_YYYYMMDD_HHMMSS.csv* when GPS time is available, otherwise *wd_bootNNNNNNNNNN.csv*. All times are UTC (GPS time). Sweden is UTC+2 in summer.
- Each network is written once per session (by BSSID). Set *LOG_ALL_SIGHTINGS* to 1 to log every sighting instead.
- Each row contains MAC, SSID, security, time, channel, frequency, RSSI and GPS position. Rows without a GPS fix leave the position fields empty.
- Press **STOP** before removing the SD card. This closes the file safely.
- The file can be imported directly into WiGLE (wigle.net) for upload.

7. Settings in the code

The settings are constants at the top of the sketch. Change them and upload again.

Constant	Default	Purpose
SCAN_MS_PER_CH	100	Time per channel in ms. 200 finds more networks but takes longer. 60 is the fastest.
TOUCH_Z_MIN	1000	Minimum touch pressure. Raise it if the screen reacts to nothing, lower it if taps are not detected.
LOG_ALL_SIGHTINGS	0	0 = one row per network, 1 = a row every scan
GPS_BAUD	9600	Baud rate of the GPS module
GPS_DEBUG_RAW	0	1 = print every raw NMEA sentence to Serial Monitor. Use for testing only.
DISPLAY_IPS / DISPLAY_INVERT	0 / 0	Change only if colours look wrong

8. Troubleshooting

Problem	What to try
Touch does not respond	Run calibration again with [CAL]. Check the serial output: each tap prints raw x, y and z.
Screen flickers	Make sure the latest sketch is uploaded. Flicker should be minimal.
SD: SD CARD MISSING	Format the card as FAT32, reseat it, and check the SD wiring. The program retries every 10 s.
GPS shows NO DATA / NO NMEA	Check TX wiring to GPIO8, 9600 baud, and 3.3 V power. Set GPS_DEBUG_RAW to 1 to see the sentences.
GPS has data but no fix	Normal indoors. Move outdoors with clear sky view and wait 2-5 minutes. The module LED blinks when it has a fix. Check the antenna is connected.
Random PSRAM error in the log	Set Tools → PSRAM to Disabled.
No serial output	Set Tools → USB CDC On Boot to Enabled, then open the Serial Monitor at 115200 baud.

Use wardriving only where you are allowed to. Logged data can contain network names and locations. Handle it responsibly.